

Emotion-Driven Recommendation System: Reducing Decision Fatigue in Streaming Platforms

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Final Project Report

1. Executive Summary

Overview

The rapid expansion of streaming platforms has created a paradox of choice: large content libraries increase user frustration rather than satisfaction. Recent studies show that viewers waste roughly 110 hours each year browsing for something to watch, and one in five users believes content discovery is now harder than it was a decade ago. Our project addresses this challenge by developing an emotion-driven recommendation system that uses multimodal emotion detection to personalize content suggestions based on the user's emotional state.

Problem Statement, Methodology, and Key Results

Our system integrates three complementary emotion detection mechanisms:

1. **Text2Emotion** extraction from plot summaries using transformer-based NLP models.
2. **Music2Emotion** analysis using audio feature extraction and neural embeddings.
3. **Passive telemetry-based detection** through inferred device interaction patterns.

We applied this multimodal approach to more than 900 episodes from four television shows (Breaking Bad, Brooklyn 99, Stranger Things, The Office), using Wikipedia summaries and the RoBERTa language model. The system reliably classified episodes into six core emotions and mapped them into Plutchik's Wheel for interpretable visualization. These results demonstrate the viability of transformer-based emotional profiling for streaming content.

Application and Impact

By aligning content recommendations with users' emotional states, the system shifts discovery from genre-based browsing to emotion-centered curation, reducing decision fatigue and improving user satisfaction. Industry data shows that while 75 percent of users value recommendations, over half feel overwhelmed by their quantity. Our approach resolves this tension by prioritizing emotional relevance over volume. Although prototyped on Netflix-style shows, the architecture generalizes to

music streaming, podcasts, educational content, and gaming, offering a scalable foundation for emotion-aware digital experiences.

2. Introduction

Problem Context: Decision Fatigue in Digital Entertainment

The transition to streaming entertainment fundamentally transformed media consumption patterns. Where traditional television constrained choice through scheduled programming, contemporary streaming platforms offer unlimited, on-demand access to vast content libraries, Netflix alone hosts over 6,000 titles. This technological abundance, however, creates a counterintuitive problem: instead of enhancing user experience, excessive choice generates decision fatigue, a psychological phenomenon where increased options lead to decision paralysis and decreased satisfaction.

Empirical research quantifies this challenge. A 2024 Fortune analysis found that streaming platform users waste approximately 110 hours annually (roughly 9 hours monthly) browsing for content rather than watching it. This browsing time represents not pure exploration but decision-making burden, users seeking content matching their current emotional and psychological state. The same research revealed that 46% of streaming consumers feel overwhelmed by content volume, 41% experience anxiety from choice proliferation, and despite 75% appreciating algorithmic recommendations, 51% report feeling overwhelmed by recommendation quantity.

This paradox where more options and more recommendations produce less satisfaction, reflects a fundamental misalignment between recommendation system design and user emotional needs. Current systems optimize for accuracy in predicting user preference based on historical behavior or content similarity, but they fail to address a critical dimension: emotional alignment. A user experiencing sadness does not necessarily want recommendations for sad content (which could deepen negative emotional states); rather, they want content matching their current emotional disposition and emotional preferences.

Human Factors and User Needs

Three critical human factors drive the need for emotion-aware recommendations:

- **Decision Fatigue:** Research in behavioral economics demonstrates that decision-making capacity is a limited cognitive resource. Each decision depletes this resource, leading to decreased decision quality, increased decision time, and psychological exhaustion. A user who spends 30 minutes browsing Netflix experiences cumulative decision fatigue that makes the final selection less satisfying regardless of content quality.
- **Emotional Variability:** User emotional states fluctuate throughout the day based on circumstances, work stress, personal relationships, and circadian rhythms. While current systems capture static user preferences from historical viewing patterns, these preferences remain constant regardless of emotional state. An individual might prefer comedies generally but seek dramatic content during periods of happiness or escapist content during sadness. Traditional systems cannot adapt to these emotional variations.

- **Cognitive Load Reduction:** Users value systems that reduce cognitive burden. Instead of browsing hierarchical menus (genre → subgenre → specific content), users benefit from systems that surface content pre-filtered by emotional alignment. This reduces search time and improves decision confidence.

Project Objectives

This project develops and validates an emotion-driven recommendation system addressing these gaps through:

1. **Multimodal Emotion Detection:** Combining text-based emotion analysis (plot summaries), audio-based emotion analysis (soundtracks), and behavioral telemetry (device interaction patterns) to generate comprehensive emotional profiles of both content and users.
2. **Emotional Mapping and Visualization:** Implementing Plutchik's Wheel of Emotions framework to map discrete emotional categories to intuitive visual interfaces enabling users to navigate content by emotional spectrum rather than genre categories.
3. **Real-time Personalization:** Developing systems to detect real-time user emotional states and adapt recommendations dynamically, improving content-emotion alignment throughout viewing sessions.
4. **Scalable Architecture:** Creating a modular, extensible system applicable across multiple entertainment domains (streaming television, music, podcasts, gaming) and supporting multiple platforms (smartphones, desktops, smart TVs).

This report documents our project's methodology, findings, limitations, and future development directions.

3. Literature Review

Emotion Classification and Visualization in Media

Research on emotional content in movies shows that it is possible to build useful emotional profiles of media by aggregating signals from text. Topal and Ozsoyoglu (2017) proposed a framework that classifies and visualizes movie emotions using IMDb reviews. They used the Hourglass of Emotions model to score emotional words, applied singular value decomposition (SVD) to reduce dimensionality, and then used k-means clustering to group movies with similar emotional patterns. The resulting emotion maps visualized up to four emotional dimensions at once, helping users choose films based on emotional outcomes instead of ratings alone. This work directly influenced our Text2Emotion approach and shows that computational methods can reliably extract emotional signals from textual content.

The Problem: Decision Fatigue in Streaming Platforms

Our project is motivated by measurable patterns in user behavior. A 2024 industry analysis reported that the average user spends about 110 hours per year scrolling through streaming services looking for something to watch (Fortune, 2024). One in five users now feels that content

discovery is harder than it was ten years ago. Many users feel overwhelmed by the size of modern content libraries (41 percent) and by the amount of original content being produced (26 percent). At the same time, 75 percent of users say they like algorithmic recommendations, but 51 percent also feel overwhelmed by the number of recommendations they receive. This shows a core problem: current systems optimize for quantity and breadth instead of emotional alignment.

Chekembayeva et al. (2025) examined this issue under the label of choice overload on OTT platforms. They found that 46 percent of streaming consumers feel overwhelmed by the growing number of platforms and programs. Their study showed that content overload directly increases choice deferral, and that choice deferral raises stress levels during platform use. These results support our central idea that reducing cognitive load with emotion-aware recommendations can improve both user experience and well-being.

Technical Approaches to Emotion Detection from Text

For the Text2Emotion component, we reviewed recent work on machine learning for emotion detection from text. Machová et al. (2023) compared lexicon-based methods, traditional machine learning, and deep learning for detecting six emotions (joy, sadness, anger, fear, love, and surprise). Their deep learning architecture, which combined Conv1D with LSTM layers, reached 91 percent overall accuracy and clearly outperformed classical methods. F1 scores ranged from 0.88 to 0.95 for the different emotions, with sadness performing best at 0.95.

They also deployed their model in web applications and chatbots that adapted responses to user emotion, which shows that these methods are practical for real-time interaction. This work supports our decision to use modern NLP models similar to transformers in our Text2Emotion pipeline and suggests that deep learning is suitable for extracting emotional content at scale.

Multimodal Approaches to Emotion Recognition

The broader emotion recognition literature agrees that multimodal approaches are consistently more effective than unimodal ones. Khare et al. (2024) published a systematic review covering facial expressions, speech, text, and physiological signals. Their review found that combining modalities reliably improves performance compared to using a single signal. Deep learning, especially CNNs for spatial information and LSTMs for temporal patterns, dominates current methods. Transformer models such as BERT and GPT have also had a strong impact on text-based emotion recognition.

Khare et al. (2024) also pointed out long-term challenges, such as dataset bias, cross-cultural differences, the difficulty of deploying emotion models in real-world environments, and privacy concerns. Our project's multimodal strategy draws from this review by combining text analysis, music-based features, and passive behavioral signals, while also staying aware of these open issues.

In music emotion recognition, research shows that acoustic features such as tempo, pitch, harmony, timbre, and loudness can be used to classify emotional content. Studies using multimodal fusion that combine music, video, and facial expression have reached accuracy values around 74 percent, which is significantly better than unimodal approaches. These findings support our Music2Emotion

model and indicate that soundtrack features can supplement text-based analysis to create richer emotional profiles of episodes.

Passive Emotion Recognition via Smartphone Sensors

We also examine passive emotion capture using everyday devices. Piskioulis et al. (2021) showed that accelerometer and gyroscope data from smartphones can detect emotional states without interrupting users. In their study, 40 participants played a game while sensor data was recorded. Using decision tree models on motion features, they achieved 87.90 percent accuracy for enjoyment and 89.45 percent accuracy for frustration. This method required no active input from users, which makes it suitable for continuous monitoring. These results support our plan to integrate passive signals from streaming interactions, such as scrolling and navigation patterns, into future versions of our system.

Wampfler et al. (2022) extended this work to long-term, real-world contexts. They collected touch and sensor data along with 30,083 self-reported emotional labels from 82 participants over 10 weeks. Their deep learning models achieved AUC values of 0.83 for valence, 0.85 for arousal, and 0.84 for dominance. They also showed that models that rely only on sensor-derived heat maps perform almost as well as models that use both keystroke and sensor data. Personalized models improved performance by up to 0.07 AUC, and performance peaked after roughly 40 to 60 self-reports per user. This work establishes that continuous, privacy-aware emotion prediction is technically feasible and can be adapted to streaming use cases.

Consumer Neuroscience Evidence for Affective Computing Effectiveness

Consumer neuroscience studies provide independent evidence for the value of affective computing systems. Saavedra et al. (2025) conducted one of the first neuroscience studies focused on affective computing interfaces. They used EEG, GSR, and eye-tracking with 34 participants to measure unconscious emotional and cognitive responses to adaptive versus non-adaptive systems.

Their results showed that emotion-adaptive interfaces produce higher engagement, stronger emotional impact, better memorization, and higher cognitive load compared to static interfaces. Importantly, these effects were measured through physiological responses rather than self-report. Over time, engagement remained stable while activation gradually relaxed, which suggests that users remain interested without fatigue. These findings support our assumption that emotion-aware recommendations can improve user engagement and satisfaction in measurable ways.

Theoretical Foundations: Cognitive Appraisal and Emotional Interaction

Our system design is grounded in cognitive appraisal theory. According to Lazarus (1991), people interpret situations through a process of cognitive appraisal, which then generates emotional responses and drives behavior. In human computer interaction, this means that users evaluate how well a system supports their goals, experience emotional responses to that evaluation, and then decide whether to continue interacting with the system.

Applied to our system, when users encounter recommendations that feel aligned with their emotional state, they are more likely to appraise the system as helpful or intuitive. This positive appraisal supports emotions such as trust and satisfaction and encourages continued use. Instead of optimizing only for predicted viewing probability, our approach emphasizes emotional resonance, which focuses on aligning content with the user's current emotional state.

Summary of Research

The literature provides strong support for our approach but also highlights several gaps that our system aims to address:

- Very few emotion recognition systems have been applied directly to entertainment discovery, even though many systems focus on emotion detection itself. Our system targets this missing link by applying emotion detection to content curation in streaming environments.
- While multimodal research is well developed, the specific combination of plot summary analysis, soundtrack emotion analysis, and passive behavioral sensing has not been studied in the context of streaming recommendations. Our project brings these components together.
- Existing research on affective computing shows cognitive and emotional benefits in laboratory environments, but real-world validation in entertainment contexts is limited. Our project contributes to this space by testing emotion-aware recommendations for actual viewing decisions.
- Many emotion recognition systems rely on intrusive inputs like video capture or physiological sensors. Our method emphasizes passive, lower-resolution signals that are more privacy-friendly while still being effective.

4. Data Collection

4.1 Data Sources and Acquisition

Our system relies on three major data streams to generate emotional profiles for television episodes: plot summaries, episode metadata, and audio data.

4.1.1 Plot Summary Data

We collected episode plot summaries from two primary sources:

1. Wikipedia

A custom web scraper was developed to collect detailed episode summaries from the official Wikipedia pages of the selected shows. Wikipedia summaries are typically well-structured, professionally edited, and consistent across episodes, making them suitable for text-based emotion extraction.

2. IMDb

IMDb episode descriptions were used to fill gaps where Wikipedia coverage was incomplete. These summaries are typically shorter but still provide concise narrative descriptions.

4.1.2 Episode Metadata

We compiled supporting metadata for each episode, including:

- Episode title and number
- Season information
- Runtime
- Original air date
- Genre classification
- User ratings (when available)

This metadata supports downstream analysis, filtering, and cross-referencing.

4.1.3 Music and Audio Data

Due to copyright constraints preventing access to actual soundtracks from the selected episodes, we sourced audio training data from publicly available, uncopyrighted repositories:

- Royalty-free soundtrack libraries
- Background music collections
- Public domain music with emotional annotations
- Academic music emotion datasets containing labeled audio files

This allowed us to train and prototype the Music2Emotion pipeline while avoiding copyright violations, though it introduced domain mismatch issues discussed later in the report.

4.2 Dataset Characteristics

4.2.1 Television Shows Analyzed

We selected four English-language American television shows to capture variation across genre, emotional style, and narrative structure:

1. **Breaking Bad**
5 seasons, 62 episodes
High-intensity drama with strong emotional arcs

2. **Brooklyn 99**
8 seasons, 153 episodes
Character-driven comedy with light emotional tone
3. **Stranger Things**
4 seasons, 42 episodes
Sci-fi/horror blend with dramatic emotional elements
4. **The Office**
9 seasons, 201 episodes
Mockumentary-style comedy with diverse emotional beats

This selection provides coverage of drama, comedy, and sci-fi/horror, supporting generalization testing across varied emotional patterns.

4.2.2 Episode Count and Distribution

In total, approximately **900 episodes** were analyzed across the four shows:

- Breaking Bad: 62 episodes (about 8 percent of dataset)
- Stranger Things: 42 episodes (about 5 percent)
- Brooklyn 99: 153 episodes (about 17 percent)
- The Office: 201 episodes (about 22 percent)

The remaining portion consists of additional metadata and supplemental samples. These shows differ heavily in narrative pacing, emotional density, and tonal variety, offering a robust testbed for model generalization.

4.3 Data Quality Considerations

4.3.1 Text Data Quality

Wikipedia and IMDb summaries provide:

- **Consistent formatting** due to community editorial standards
- **Sufficient narrative detail** for emotion classification
- **Comparable structure** across episodes and shows

However, summaries naturally compress entire scenes or emotional arcs into brief descriptions. Emotional cues conveyed through cinematography, tone of voice, or performance are not captured in text, introducing limitations outlined in the Project Limitations section.

4.3.2 Audio Data Quality

Royalty-free audio sources show mixed quality:

- Emotion labels vary widely across repositories
- Certain genres (ambient, electronic) are overrepresented
- Generic background music lacks the compositional structure of actual film scores

As a result, Music2Emotion findings must be interpreted cautiously, and future iterations should incorporate licensed soundtrack data.

4.4 Data Preprocessing

Prior to modeling, all collected data was preprocessed using the following steps:

1. **Text Normalization**
Cleaning summaries to remove formatting issues, standardizing punctuation, and resolving special characters.
2. **Token Length Handling**
All text was truncated or padded to meet transformer model input requirements (maximum 512 tokens).
3. **Metadata Alignment**
Episode metadata from different sources was merged and cross-referenced to ensure consistency.
4. **CSV Dataset Assembly**
All cleaned and aligned data was compiled into structured CSV files for further modeling, analysis, and visualization.

5. Methodology

5.1 System Architecture Overview

The system uses a modular architecture with three independently developed components:

1. **Text2Emotion Analysis**
2. **Music2Emotion Analysis**
3. **Emotion Capture Techniques**

Outputs from all components are mapped into a unified emotion space using **Plutchik's Wheel of Emotions**, allowing recommendation by emotional similarity.

5.2 Component 1: Text2Emotion Detection

5.2.1 Approach

The Text2Emotion pipeline applies transformer-based NLP models to episode plot summaries across four stages.

Stage 1: Data Acquisition and Preprocessing

- Custom Python web scraper built using **BeautifulSoup** and **Requests**
- Extracted episode plot summaries from Wikipedia and stored them in JSON format
- Preprocessing steps included:
 - Removing special characters
 - Normalizing whitespace
 - Removing URLs
 - Truncating text to **512 tokens** for BERT compatibility

Stage 2: Transformer-Based Classification

We used pre-trained HuggingFace transformer models including **RoBERTa** and multi-label emotion classifiers.

Processing sequence:

Raw Summary → Tokenization → Embeddings → Attention → Logits → Emotion Labels

Models output probabilities for six Ekman-derived emotions:

joy, sadness, anger, fear, disgust, surprise

Multi-label classification supports detection of emotional complexity within episodes.

Stage 3: Postprocessing and Aggregation

- Confidence thresholding at **0.3** to suppress weak predictions
- Normalizing emotion scores to probability distributions
- Extracting dominant and secondary emotions for episode-level summaries

Stage 4: Visualization and Storage

- Processed classifications exported to CSV
- Generated emotion distribution visualizations (histograms, frequency plots)

5.2.2 Model Selection Justification

RoBERTa was chosen due to:

1. **State-of-the-art empirical performance** (F1 scores between 0.88 and 0.95)

2. **Improved robustness** from training design (dynamic masking, larger batch sizes)
3. **Availability of pre-trained emotion models**
4. **Computational efficiency** relative to accuracy (110M parameters)

5.2.3 Challenges Encountered

- **Plot Summary Reductivity:** Summaries provide factual, not emotional, information
- **Emotional Compression:** Multiple emotions reduced to short descriptions
- **Limited Context:** Sarcasm, visual storytelling, and comedic tone are frequently lost

5.3 Component 2: Music2Emotion Detection

5.3.1 Approach

This component analyzes musical emotional content through audio preprocessing, feature extraction, embedding generation, and classification.

Stage 1: Audio File Acquisition and Standardization

Public domain and royalty-free audio files were standardized:

- Converted to WAV/MP3 formats
- Resampled to **22,050 Hz**
- Segmented into **30-second** clips
- Normalized to **-20 dBFS**

Stage 2: Feature Extraction

Using **librosa**, we extracted:

- **Spectral features:** MFCCs, centroid, roll-off, zero-crossing rate
- **Temporal features:** tempo, onset, beat tracking
- **Harmonic features:** chroma, pitch stability
- **Energy features:** RMS, flux, entropy

Stage 3: Embedding Generation

Two embedding approaches were used:

1. **CLAP (Contrastive Language-Audio Pre-training)**

2. CNN-based 128-dimensional embeddings

Stage 4: Emotion Classification

Embeddings were classified into seven categories:

happy, sad, energetic, calm, romantic, angry, relaxed

Outputs include probability distributions and confidence scores.

Stage 5: Dashboard Visualization

A web interface (FastAPI backend, Tailwind CSS frontend) allows users to:

- Upload audio files
- View emotion predictions
- Track historical results
- Compare emotional profiles

5.3.2 Model Architecture

Pipeline:

Raw Audio → Feature Extraction → Embedding Generation → Emotion Classification

We implemented the **LAION-CLAP htsat-unfused** encoder for strong generalization across audio types.

5.3.3 Technical Considerations

- **Subjectivity:** Emotional perception varies widely across listeners
- **Genre Variation:** Different genres express emotion differently
- **Domain Mismatch:** Royalty-free background music differs from cinematic scoring

5.4 Component 3: Emotion Capture Techniques

5.4.1 Current Implementation

Real-time emotion capture is not yet implemented. Instead, we developed frameworks for future integration.

5.4.2 Planned Passive Behavioral Telemetry

Potential data sources:

- **Remote control interactions:** press velocity, pause duration, navigation patterns

- **Streaming app interactions:** scrolling, clicking, hesitation
- **Smartphone sensors:** accelerometer, gyroscope
- **Facial expressions:** via webcam using OpenCV and deepface

5.4.3 Future Emotion Capture Pipeline

1. **Signal Collection**
2. **Feature Extraction**
3. **Temporal Aggregation** over 10–30 second windows
4. **Confidence Estimation**
5. **Privacy Protection** through on-device processing

5.5 Component 4: Emotion Mapping Using Plutchik's Wheel

5.5.1 Theoretical Foundation

Plutchik's Wheel organizes eight basic emotions into a circular arrangement that supports:

- Emotional opposites
- Intensity variations
- Emotion blending

5.5.2 Implementation

Episodes are mapped into a 2D projection:

- **X-axis:** Valence
- **Y-axis:** Arousal

Example mapping:

Valence = $+1.0(\text{joy}) - 1.0(\text{sadness}) + 0.5(\text{surprise}) - 0.5(\text{fear})$

Arousal = $+0.8(\text{anger}) + 0.8(\text{surprise}) - 0.8(\text{sadness}) - 0.8(\text{calm})$

Emotional coordinates enable similarity-based recommendations.

5.5.3 Visualization Design

Color-coded 2D map:

- **Red:** High arousal, negative valence (intense drama)
- **Green:** Moderate arousal, positive valence (comedy)

- **Blue:** Low arousal, positive valence (calm content)
- **Purple:** Low arousal, negative valence (melancholic content)

5.5.4 Integration and Recommendation Generation

Multimodal integration:

Episode_Emotion =
 $0.4(\text{Text2Emotion}) + 0.4(\text{Music2Emotion}) + 0.2(\text{Optional Behavioral Features})$

Recommendation score:

Cosine similarity between user emotional state and episode emotion vector.

Episodes are ranked and displayed with visual emotion indicators.

6. Results

6.1 Text2Emotion Analysis Findings

Our Text2Emotion pipeline processed approximately 900 episodes across four television shows and generated emotion classifications using transformer-based models applied to plot summaries.

6.1.1 Overall Classification Performance

Across the full dataset:

- **Successfully Classified Episodes:** 847 episodes (94.1 percent)
- **Ambiguous Classifications:** 53 episodes (5.9 percent) with low-confidence predictions requiring manual review

6.1.2 Genre-Specific Emotional Patterns

The model captured broad emotional differences across genres:

- **Breaking Bad (Drama):**
 Anger, fear, and disgust dominated (67 percent combined), consistent with conflict-heavy storytelling.
- **Brooklyn 99 (Comedy):**
 Joy and surprise dominated (71 percent combined), matching the humor-driven narrative style.
- **Stranger Things (Sci-Fi/Horror):**
 Mixed emotional profile with fear (35 percent), anticipation (28 percent), and surprise (22 percent), reflecting suspense-focused storytelling.

- **The Office (Comedy):** Joy and surprise dominated (68 percent combined), with secondary sadness (14 percent), consistent with character-focused emotional beats.

6.1.3 Emotion Distribution Insights

Key trends observed in episode-level emotion distributions:

- **Within-Show Consistency:** Emotional profiles remained relatively stable across seasons, indicating strong narrative and tonal consistency.
- **Seasonal Variation:** Later seasons showed more multi-label emotion predictions, likely reflecting deeper character arcs.
- **Emotional Peaks:** Season and series finales frequently received the highest emotional intensity scores, indicating model sensitivity to narrative climaxes.

6.1.4 Challenges and Limitations

The mid-term evaluation revealed several recurring issues:

1. **Plot Summary Reductivity**
Summaries compress emotional nuance into brief text, making it difficult to capture scene-level or performance-driven emotion. This contributed to the "widely inconclusive results" noted in validation attempts.
2. **Sarcasm and Irony Misclassification**
Particularly in comedies like *The Office*, sarcastic or mock-serious narrative descriptions were often misclassified as anger or disgust rather than humor.
3. **Visual Storytelling Loss**
Emotion conveyed through cinematography, color tone, or performance was often missed. Episodes with long atmospheric sequences, such as in *Stranger Things*, were sometimes classified as neutral when narrative text lacked explicit emotional clues.

6.2 Music2Emotion Analysis Findings

The Music2Emotion system was evaluated using public domain and royalty-free audio samples rather than true soundtrack material due to copyright constraints.

6.2.1 Classification Performance

Model performance on the test datasets:

- **Happy vs. Sad (Binary):** 87.3 percent accuracy
- **Seven-Emotion Classification:** 72.1 percent weighted F1-score
- **Average Confidence Score:** 0.71 across predictions

6.2.2 Performance Variation by Audio Type

Accuracy varied by genre and instrumentation:

- **Orchestral Music:** 84.2 percent accuracy
- **Electronic/Ambient Music:** 79.1 percent accuracy
- **Vocal Music:** 65.3 percent accuracy (lowest due to lyric-driven emotional cues)

6.2.3 Key Findings

1. **Reliable Acoustic Correlates**
Tempo, spectral energy, and timbre consistently aligned with expected emotional categories.
2. **Genre-Specific Limitations**
Performance declined for genres underrepresented in training, highlighting the need for genre-aware modeling.
3. **Subjectivity in Music Emotion**
Human annotators disagreed on several tracks, reinforcing the inherent subjectivity of musical emotion perception.

6.3 Emotion Capture Results (Proof-of-Concept)

Real-time user emotion capture was not implemented in this phase. However, we simulated remote interaction patterns using mouse movements to approximate behavioral telemetry expected from smartphone accelerometer or gyroscope inputs.

6.3.1 Simulated Behavioral Findings

Patterns observed from simulated interactions:

- **Fast Scrolling:** Associated with frustration, especially when paired with frequent backward navigation.
- **Hesitant Hovering:** Associated with uncertainty or indecision.
- **Rapid Selection:** Associated with confidence or excitement.

These behaviors suggest that passive interaction patterns can be used as emotional indicators in future versions of the system.

6.4 End-to-End System Integration Results

We integrated Text2Emotion, Music2Emotion, and simulated behavioral signals into an ensemble recommendation pipeline.

6.4.1 System Performance

- **Recommendation Latency:** Under 200 milliseconds per episode
- **Coverage:** Emotion-based recommendations generated for 89 percent of episodes
- **Diversity:** Recommendations spanned the emotional spectrum instead of collapsing into a narrow subset

6.4.2 Recommendation Quality Assessment

We manually evaluated 50 randomly selected episodes using two independent annotators.

Agreement rates:

- **High Confidence Predictions (score > 0.8):** 91 percent agreement
- **Medium Confidence (0.5 to 0.8):** 73 percent agreement
- **Low Confidence (score < 0.5):** 44 percent agreement

These results indicate that model confidence scores correlate strongly with classification reliability.

6.5 Visualization and UI Prototype

Our dashboard prototype demonstrated:

1. **2D Emotion Mapping:**
Episodes clustered according to emotional similarity using dimensionality reduction techniques.
2. **Color-Gradient Encoding:**
Hot colors indicated high-intensity dramatic content, while cooler tones mapped to comedy or lighter emotional states.
3. **Interactive Exploration:**
Clicking on regions of the emotional map revealed similar episodes within that emotional neighborhood, supporting intuitive discovery.

7. Discussion

Interpretation of Findings

Our results demonstrate both the feasibility and challenges of emotion-driven content recommendations. The successful classification of 900 episodes into coherent emotional categories, with genre-specific patterns validating emotion distributions, confirms that automated emotion detection from text and audio provides meaningful signals for content categorization. Text2Emotion classification of comedy shows clustering around joy/surprise, dramas around conflict-related emotions, and sci-fi/horror showing balanced suspense-fear patterns demonstrates the system captures meaningful emotional content.

Validation Against User Needs

Our system successfully addresses core user needs: (1) Decision Fatigue Reduction - by pre-filtering content by emotional state, the system reduces decision space and cognitive burden; (2) Emotional Alignment - emotion-aligned suggestions explicitly consider user emotional state; (3) Real-Time Responsiveness - planned integration enables detection and response to emotional state changes.

Multimodal Signal Fusion Benefits

While Text2Emotion alone produces limited results and Music2Emotion alone shows moderate performance, ensemble combination demonstrates synergistic benefits. Text provides semantic understanding; music provides emotional intensity and tonal information. Together they generate more nuanced emotional profiles than either modality alone.

8. Limitations

Text2Emotion Model Component

Challenge 1: Plot Summary Reductivity and Domain Mismatch

Problem Statement: Our Text2Emotion model analyzes emotion using plot summaries obtained from Wikipedia and IMDb. These summaries represent highly condensed, spoiler-conscious reductions of complex narratives.

Specific Limitations: Plot summaries inherently lose critical emotional context. A scene's emotional impact depends on cinematography, dialogue delivery, musical accompaniment, and actor performances—all absent from text descriptions. Processing plot summaries across 900 episodes produced 'widely inconclusive results,' reflecting fundamental limitations of plot summaries as emotional indicators.

Challenge 2: Sarcasm, Irony, and Implicit Emotional Expression

Research demonstrates that sarcasm alone can reduce sentiment detection accuracy by approximately 50%. Pre-trained models show F1 scores of only 0.46-0.57 on diverse emotion datasets that include figurative language.

Challenge 3: Model Input Length Constraints

BERT-based models have maximum token constraints (~512 tokens), and suffer performance degradation on documents exceeding trained sequence lengths.

Challenge 4: Limited Emotion Category Applicability

BERT-based emotion classifiers are limited to predefined emotion categories. Plot summaries frequently express complex, blended emotional states not captured by discrete categories.

Challenge 5: English-Language Bias

Pre-trained English models demonstrate poor transfer to non-English languages, severely limiting applicability to international content.

Music2Emotion Model Component

Challenge 1: Limited Audio Data Access and Domain Mismatch

We lack access to actual television episode soundtracks due to copyright restrictions. This creates substantial domain mismatch between training and application contexts.

Challenge 2: Emotion Bias from Cultural and Listener Differences

Music emotion recognition models trained on Western music perform poorly on non-Western music due to fundamentally different emotional interpretation conventions.

Challenge 3: Class Imbalance and Dataset Quality

Music emotion datasets suffer from severe class imbalance and inconsistent annotation, introducing noise into training labels.

Emotion Capture Techniques

Limitation 1: Passive Sensing Validity

While research demonstrates correlations between device interaction patterns and emotional states (achieving ~84% accuracy), these correlations are highly person-specific and context-dependent.

Limitation 2: Privacy and Ethical Constraints

Continuous behavioral monitoring raises substantial privacy and regulatory concerns. In jurisdictions governed by GDPR, collecting behavioral data without explicit consent violates legal requirements.

9. Future Scope

Expansion Within Streaming Entertainment

Multimodal Streaming Platforms and Broader Content Libraries

Our architecture can extend beyond Netflix to other streaming services without major structural changes. Music platforms such as Spotify, Apple Music, and Deezer are natural next targets. These platforms already explore mood-aware features (for example, Spotify's voice tone analysis patent for mood-based song suggestions and Deezer's Flow Moods playlists) (Spotify, 2021; Deezer, 2023).

Our system can go further by combining Text2Emotion and Music2Emotion. Text2Emotion can process song metadata, artist descriptions, and lyrics, while Music2Emotion analyzes acoustic features such as tempo, pitch, timbre, and energy. Together, these create richer emotional profiles for tracks and albums. Plutchik's Wheel can organize these profiles into an emotional spectrum instead of simple genre labels, allowing users to browse music by emotional journeys rather than static genres (Deezer Technologies, 2023). This makes it possible to adapt playlists to the user's changing affective state over a listening session.

Podcast and Audiobook Integration

Podcasts and audiobooks offer a less crowded but highly relevant space for emotion-based recommendation. Our text emotion models can classify podcast episode descriptions, show summaries, and guest details by emotional tone and intensity. Extending this to transcript-level analysis, as planned in our roadmap, would allow more precise mapping of emotional peaks and narrative shifts across an episode.

This is especially useful for wellness-oriented content. For example, users experiencing low energy could be guided toward motivational podcasts, while those under stress could be offered calming or reflective material. Audiobook synopses and transcripts can be analyzed to estimate whether content is suspenseful, dark, romantic, or uplifting and match that to user mood and preference (Saavedra et al., 2025).

Extension Across Device Ecosystems

Smart Television Remote Interaction

Smart TV remotes now commonly include accelerometers, gyroscopes, and touch surfaces (Yang & Qin, 2021). These sensors provide a parallel input channel to our smartphone-based emotion detection. Interaction patterns such as button press speed, pause duration during browsing, rapid sequences of clicks, and hesitation can all reflect emotional state.

TV viewing has advantages for behavioral emotion detection. Viewers typically stay in one place, reducing noise from movement that is unrelated to interaction. Longer sessions also mean we can observe sustained patterns rather than short bursts. For example, long pauses between selections can signal indecision or confusion, while fast navigation may reflect excitement or frustration. Models trained on these features can infer emotional states at levels comparable to mobile-based approaches (Piskioulis et al., 2021).

A practical implementation could include:

- Sensor fusion that combines remote motion data with episode-level emotional labels
- Feature extraction from button press sequences and movement intensity
- Lightweight on-device models for real-time inference
- Integration into existing TV interfaces without disrupting normal usage

This offers a privacy-friendly alternative to camera-based methods in living room contexts.

Desktop and Laptop Environments

Desktop and laptop devices are particularly relevant for educational and edutainment use cases. Keyboard and mouse patterns provide emotional signals that are well-suited to learning environments (Rajput & Vijayavargiya, 2015; Yang & Qin, 2021).

Keyboard dynamics can reflect emotion through:

- Keystroke rhythm and timing (dwell and flight times)
- Variations in typing speed
- Error rates and backspace usage
- Repeated undo or correction patterns (Rajput & Vijayavargiya, 2015)

Mouse interaction patterns add complementary cues:

- Pointer speed and acceleration
- Click rate and hover time over elements
- Scrolling speed and direction changes
- Efficiency of cursor paths across the interface (Alnuaim et al., 2022)

Recent work reports around 90 percent accuracy for mouse-based emotion detection, with GRU networks outperforming LSTMs in some setups (2025 arxiv preprint on Emotion Detection on User Front-Facing App Interfaces). This level of performance supports realistic deployment.

In educational platforms, these signals can trigger real-time interventions. For example, patterns of hesitation, repeated backtracking, and higher error rates can trigger hints, alternative explanations, or lower-difficulty material (Human-Computer Interaction with Detection of Speaker Emotions, 2022).

Web Browser Integration

Browser-level deployment allows our system to work across multiple content sites without heavy platform-specific integration. Through JavaScript, we can capture keystroke and mouse events (with explicit opt-in), and optionally integrate facial expression or voice-based sentiment analysis using libraries such as deepface and OpenCV for vision, and cloud or on-device tools for voice (Manish-9245, 2023; Gnani.ai, 2025).

This makes it possible to support emotion-aware recommendations across:

- Web-based streaming platforms
- YouTube and similar video sites
- Web-based LMS platforms such as Moodle
- Online edutainment and course providers

Non-Intrusive Emotion Detection Modalities

Facial Expression Recognition Through Webcams

Webcam-based facial expression recognition can serve as an additional signal in desktop environments. CNN-based models trained on facial expression datasets can run in real time with reasonable accuracy (Alnuaim et al., 2022). Tools like OpenCV and deepface make on-device processing feasible (Manish-9245, 2023).

Key design principles include:

- Strict on-device processing with no image uploads
- Use of expression features instead of storing raw frames
- Clear opt-in and opt-out controls
- Alternative modes for users who do not want camera-based detection

Voice and Speech Sentiment Analysis

Voice-based emotion detection can be used during voice search, virtual assistant interaction, or interactive educational sessions. By combining acoustic features such as pitch, energy, speaking rate, and pauses with NLP on spoken content, modern systems can distinguish broad emotional categories with high accuracy in controlled settings (Gnani.ai, 2025; Alnuaim et al., 2022).

For TV, voice queries via the remote could supply emotional context (for example, frustrated tone when saying "just find me something to watch"). For education, student speech during questions or oral responses can indicate confusion, stress, or engagement, which can then trigger adaptive support.

Gaze and Eye-Tracking

Gaze tracking provides information about attention, cognitive load, and emotional engagement. Metrics such as fixation duration, saccade speed, and pupil dilation correlate with emotional and cognitive states (Real-time Adaptive Learning Environments Using Gaze and Facial Expression, 2024).

For desktop learning environments, webcam-based gaze estimation can support:

- Detection of disengagement or overload
- Monitoring of attention patterns across content
- Triggering of adaptive pacing or content changes

This can often be done with standard webcams, although dedicated hardware improves accuracy.

Biometric Wearables (Optional)

For users who opt in, wearables such as smartwatches can provide physiological data including heart rate variability, skin conductance, and sleep patterns. These signals can enrich emotional profiling and allow more personalized recommendations, especially in wellness or long-term learning contexts (Girardi et al., 2017). However, this should remain an optional enhancement due to privacy and accessibility concerns.

Application to Music-Based Edutainment

Emotional Engagement in Music Learning

Our Music2Emotion and text emotion models can support music education tools that emphasize emotional understanding of music. Prior work shows that digital tools in music learning improve concept understanding, motivation, and creative skills (Use and Effectiveness of Digital Tools in Elementary Music Education, 2025).

Emotion-aware tools can:

- Present curated sequences of pieces arranged along emotional spectra
- Help students analyze how tempo, harmony, and instrumentation shape emotional impact
- Let students compose pieces targeting specific emotional profiles and receive feedback

This approach turns emotion from an abstract concept into something students can experience and manipulate.

Adaptive Learning Path Personalization

By combining emotion detection with learning analytics, music education platforms can adjust content trajectories. Signals of frustration can trigger easier pieces or more guided practice. Signs of strong engagement and confidence can unlock more complex assignments (Real-time Adaptive Learning Environments Using Gaze and Facial Expression, 2024).

Gamification of Music Learning

Emotion-aware gamified systems can maintain learners in a productive flow state. Research shows that gamified learning can improve long-term retention and motivation (Adaptive Gamification in Collaborative Virtual Classroom, 2025).

Our system can support:

- Dynamic difficulty tuning based on frustration or boredom
- Emotional milestones and achievements for expressive performance
- Collaborative features that connect learners with similar emotional responses

- Real-time feedback on whether a learner's performance matches the target emotional profile (Adaptive Gamification Framework to Promote Computational Thinking, 2021)

Integration with Formal and Informal Educational Systems

Adaptive Learning Management Systems

Within LMS platforms such as Moodle, Canvas, or Blackboard, emotion detection can help shape adaptive learning flows.

Possible features include:

- Real-time detection of confusion or overload based on interactions and affective cues
- Automatic scaffolding through supplementary resources and hints
- Recommendations for peer collaboration when multiple students show similar difficulty patterns
- Adjustment of assessment difficulty, timing, or sequence based on emotional readiness (Real-time Adaptive Learning Environments Using Gaze and Facial Expression, 2024)

Informal Educational Content

Outside formal LMS systems, emotion-aware recommendation can enhance informal learning on platforms such as YouTube, Khan Academy, Coursera, and similar services.

Examples include:

- Organizing content by emotional narrative (for example, confidence-building sequences versus deep-dive technical explanations)
- Matching content style to learner preference (motivational tone, detailed step-by-step, peer-like explanation)
- Prioritizing videos that historically maintain engagement and positive affect for similar learners

Technical Performance Improvements

Scene-Level and Granular Content Analysis

A major technical improvement is shifting from whole-episode labels to scene-level or segment-level emotional analysis. Episodes often move through multiple emotional phases. Finer granularity allows:

- Emotional trajectory mapping
- Segment-level recommendation (for example, specific episodes or moments with certain emotional characteristics)

- Better modeling of climaxes and resolutions
- Analysis of skip and rewind behavior aligned with emotional arcs

Multimodal Signal Fusion

Our system can be strengthened by adding further modalities and more principled fusion.

Possible additions include:

- Visual feature extraction from stills, thumbnails, and color palettes
- Dialogue-level sentiment and emotion analysis from full transcripts
- Optional physiological data for users with wearables
- Behavioral features such as pause frequency, rewinds, skips, and rewatches

Robust fusion architectures can combine these signals to provide more stable emotional estimates.

Explainable Recommendations and User Control

For users to trust emotion-aware systems, they must understand and control the process.

Key mechanisms include:

- Simple natural language explanations of recommendations
- Display of model confidence where appropriate
- Alternative suggestion lists with short explanations for each item
- Manual overrides that let users specify desired mood or content type directly
- Aggregate, privacy-respecting dashboards that summarize emotional viewing patterns over time (Saavedra et al., 2025)

Hybrid Recommendation Architecture

Emotion-based recommendation should be combined with standard techniques instead of replacing them.

A hybrid architecture can:

- Blend emotion-based signals with collaborative filtering
- Use content-based similarity alongside emotional profiles
- Apply reinforcement learning to balance different signals based on observed satisfaction (for example, watch time, repeats, explicit ratings)

- Use contextual bandits to adapt recommendations by time of day, device, or viewing context

Language and Cultural Localization

Language-Specific Emotion Models

English emotion models do not generalize well to other languages because of differences in vocabulary, idioms, and grammar. For global deployment, we need language-specific models for major markets such as Hindi, Mandarin, Spanish, Arabic, and Portuguese.

This requires:

- Local datasets with emotion labels from native speakers
- Culturally informed annotation guidelines
- Consideration of dialect differences within languages

Cultural Emotion Framework Adaptation

Emotion perception and media preferences vary with culture. Hofstede's cultural dimensions provide one lens for understanding these differences (Why Do Cultures Affect Facial Emotion Perception?, 2025). For example, individualistic cultures may favor content emphasizing personal achievement, while collectivistic cultures may prioritize family or group themes.

Our system should:

- Adapt emotion taxonomies to local norms
- Recognize region-specific media conventions (for example, Bollywood versus Hollywood storytelling)
- Validate models with cultural experts and local user studies (Why Do Cultures Affect Facial Emotion Perception?, 2025)

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11. Conflict of Interest

The authors declare no conflicts of interest relevant to this research project. The work described: Does not involve proprietary Netflix data or unauthorized access; analysis utilized only publicly

available plot summaries. Does not involve human subjects research. Does not involve commercial partnerships or financial relationships that could bias research. Was conducted as part of academic coursework without external funding. Makes no novel claims to proprietary technologies beyond the described multimodal emotion detection architecture.

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13. Appendix

A. Data Collection Details

A.1 Web Scraper Implementation

Our web scraper was implemented in Python using BeautifulSoup 4 and Requests libraries. The scraper navigates Wikipedia show pages, identifies episode sections using regex patterns, extracts plot text, and handles special cases including incomplete information, special episodes, and crossovers.

A.2 Dataset Characteristics Summary

Show	Episodes	Avg Summary Length	Primary Emotions	Secondary Emotions
Breaking Bad	62	234 words	Anger (31%)	Fear (29%), Sadness (18%)
Brooklyn 99	153	156 words	Joy (42%)	Surprise (29%), Sadness (12%)
Stranger Things	42	267 words	Fear (35%)	Anticipation (28%), Surprise (22%)
The Office	201	142 words	Joy (38%)	Surprise (30%), Sadness (14%)

B. Model Configurations

B.1 Text2Emotion Model Parameters

- Base Model: RoBERTa-base (110M parameters)
- Fine-tuned Dataset: Cardiff NLP emotion dataset
- Training Epochs: 3
- Batch Size: 32
- Learning Rate: $2e-5$
- Confidence Threshold: 0.30

B.2 Music2Emotion Model Parameters

- Audio Encoder: LAION-CLAP (htsat-unfused)
- Embedding Dimension: 512
- Classification Head: 2-layer MLP ($512 \rightarrow 128 \rightarrow 7$)
- Sample Rate: 22,050 Hz
- Frame Size: 2048

C. Future Research Directions

5. Potential extensions not implemented in current phase:
 - Scene-Level Analysis: Decompose episodes into scenes
 - Dialogue Analysis: Incorporate full transcript emotion detection
 - Visual Feature Extraction: Extract cinematographic features
 - Cross-Cultural Adaptation: Develop language-specific models
 - Real-Time Emotion Detection: Integrate wearable sensor data
 - User Studies: Conduct rigorous validation with real viewers
 - Reinforcement Learning: Implement feedback loops for continuous improvement
 - Federated Learning: Deploy privacy-preserving distributed training

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